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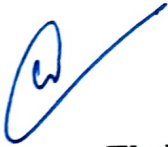
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This is to certify that the Project Report entitled “Skill Matcher” which is submitted by Animay Patel, Aman, Swapnil Suman, Shivam Raj in partial fulfillment of the requirement for the award of degree B. Tech. in the department of Computer Science & Engineering of KIET Group of Institutions, Delhi NCR affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.



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Paper Title: Automated Skill Extraction and Resume Matching Using BERT-Based NLP Model

Author(s): Animay Patel, Swapnil Suman, Shivam Raj, Aman, Gagan Thakral, Malay Kumar



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A
Project Report
on
Skill Matcher
submitted as partial fulfilment for the award of
BACHELOR OF TECHNOLOGY
DEGREE

SESSION 2024-25
in
Computer Science & Engineering

by
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May, 2025

DECLARATION

We hereby declare that this submission is our work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgement has been made in the text.

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ABSTRACT

In the present hiring environment, manually screening resumes and matching candidate qualifications with job criteria is time-consuming and error-prone. In order to achieve two primary objectives, this study provides a comprehensive system that employs a refined BERT-based model: Automatically extracting skills from job descriptions and resumes, and thoroughly analysing data on skills trends across multiple dimensions, such as years of experience (YOE), job position, salary, and country. Additionally, advanced text similarity metrics and adversarial training techniques are being researched to improve assessment and resilience. Our data analytics module displays statistically significant market trends, and experimental results demonstrate that the BERT-based model achieves excellent accuracy in semantic matching and named entity recognition (NER).

The integration of adversarial techniques and contemporary metrics, including BERTScore, significantly improves system reliability and presents new avenues for research. Additionally, the project helps to alleviate inefficiencies in hiring by filling gaps in skill alignment and providing actionable insights for strategic decision-making. Through the use of state-of-the-art natural language processing (NLP) and machine learning (ML) methods, the system provides flexibility to adapt to changing market demands. This book offers a solid foundation for applying data-driven perspectives in hiring, empowering both recruiters and job candidates. Future enhancements may include integrating real-time market analysis and applying the framework to behavioral assessments.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
BERT	Bidirectional Encoder Representations from Transformers
BIO	Beginning-Inside-Outside (Tagging Scheme for Named Entity Recognition)
CERT	Certification
CPU	Central Processing Unit
DL	Deep Learning
ESCO	European Skills, Competences, Qualifications and Occupations
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
HR	Human Resources
IT	Information Technology
JSON	JavaScript Object Notation
LLM	Large Language Model
ML	Machine Learning
NER	Named Entity Recognition

NLP	Natural Language Processing
O*NET	Occupational Information Network (U.S. Department of Labor)
PII	Personally Identifiable Information
POS	Part-of-Speech
ROC	Receiver Operating Characteristic
SBERT	Sentence-BERT
SQL	Structured Query Language
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
UI	User Interface
UX	User Experience
YOE	Years of Experience

CHAPTER 1

INTRODUCTION

Conventional recruitment methods have historically relied on human resume evaluation and rule-based keyword matching. These approaches frequently neglect the subtleties of natural language, resulting in inadequate candidate selection. Recent advancements in deep learning, particularly the emergence of models based on transformers such as bert (bidirectional encoders) reflect have transformed natural language processing (nlp) by offering sophisticated, context-sensitive embeddings. The increase in internet job listings has produced extensive unstructured data, the examination of which can provide insights into regional, industry, and experience trends in skill demands. This research seeks to automate the processes of skill extraction and matching, as well as the following analysis of market trends. skill extraction and matching automate the identification of skill-related items from job descriptions and resumes utilising a fine-tuned bert model. develop a semantic matching algorithm that measures the alignment between candidate skills and job specifications.

Consolidate and examine job posting data to comprehend the demand for talents across various geographies, salary brackets, roles, and years of experience (yoe). It incorporates cutting-edge text similarity metrics and adversarial training methodologies to assess and enhance model robustness and dependability.

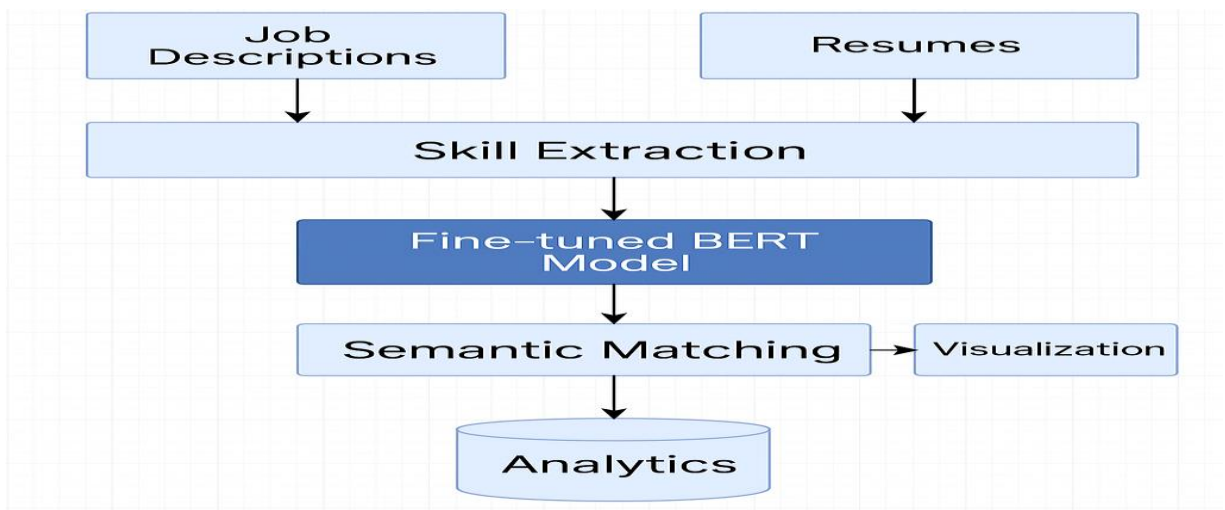


Fig 1.1: Architecture diagram of the Skill Matcher

1.1 BACKGROUND

The job market has seen rapid digital transformation in recent years. With the proliferation of online job portals and digital resumes, recruiters face the challenge of efficiently screening numerous resumes to match them with job descriptions. This process is often manual, time-consuming, and prone to bias or errors.

In addition, job seekers frequently struggle to identify roles that align with their skills due to inadequate tools for understanding job requirements and industry trends. Advancements in Natural Language Processing (NLP) and machine learning, such as BERT-based models, offer an opportunity to automate and enhance this process.

The Skill Matcher project leverages these technologies to streamline resume screening, extract skills from unstructured text, and provide actionable insights for better decision-making.

1.2 PROBLEM STATEMENT

The hiring process is greatly hindered by inefficiencies and variability in resume screening and job matching. Recruiters tend to take considerable time sorting through resumes manually, thus introducing greater opportunities for overlooking potential matches because of information overload. Standardization of the presentation of skills in resumes and job descriptions complicates the process further, with the result that it becomes tough to spot and match relevant qualifications. Besides, conventional processes have no ability to respond to the ever-changing environment of job markets, resulting in delays, inaccuracies, and frustration for candidates and recruiters alike.

Another missing link is a lack of exhaustive analytics tools with which to look at market patterns, including market trends for such things as hot skills, pay benchmarks, and geographic opportunity. Without these kinds of tools, companies are faced with workforce planning difficulties, and candidates are missing out on best-fit opportunities. The manual nature of current processes compounds these problems, leading to misaligned placements and ineffective hiring cycles. A scalable, automated solution is required to solve these problems, decrease errors, and establish a more efficient, data-driven recruitment system for all stakeholders.

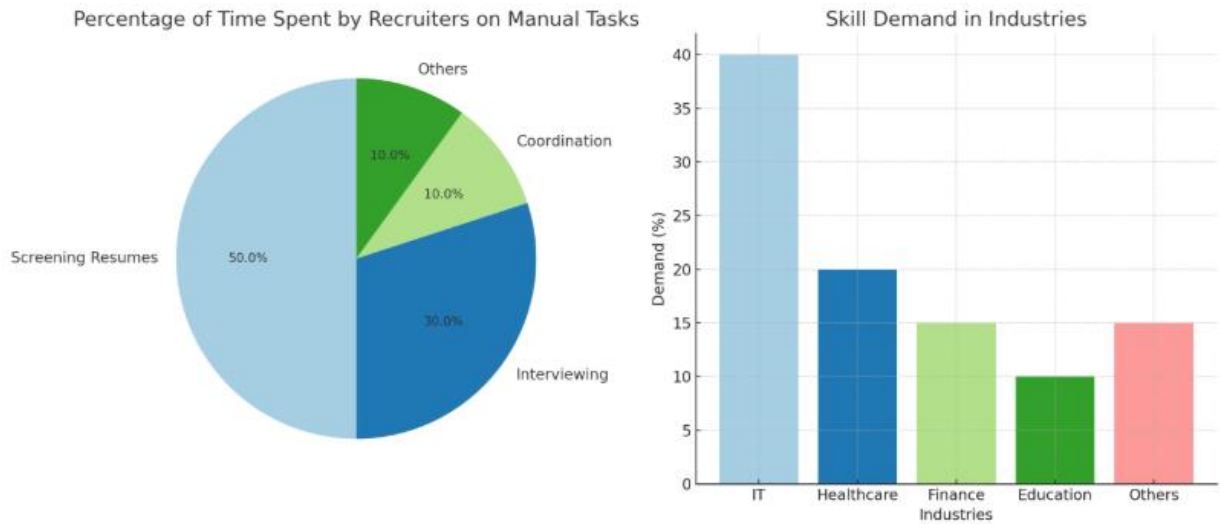


Fig 1.2: Skill Demands and Time Spend on Manual Tasks (GFG)

Recruitment and job matching processes face several challenges:

- 1. Inefficiency in Resume Screening**

Recruiters spend significant time analyzing resumes, often missing potential matches due to information overload.

- 2. Lack of Standardization in Skill Identification**

Resumes and job descriptions use varied formats, leading to inconsistencies in identifying relevant skills.

- 3. Absence of Comprehensive Analytics**

Organizations and job seekers lack tools to analyze market trends, such as salary expectations, demand for skills, and opportunities by location.

- 4. Susceptibility to Errors**

Manual methods of matching resumes to job descriptions often result in inaccuracies or missed opportunities.

Table 1.1: Comparison Between traditional and Skill Matcher

Feature	Traditional System	Skill Matcher
Efficiency	Low	High
Accuracy	Moderate	High
Time-to-Hire	High	Low
Scalability	Low	High

1.3 OBJECTIVE

The Skill Matcher project has the objective of solving key recruitment challenges using sophisticated AI methods and data analysis. Through automating labor-intensive processes, semantic enhancement, and actionable market intelligence, the project is working towards revolutionizing hiring processes. It aims to close the gap between job seekers and recruiters, allowing for faster and smarter decision-making.

The Skill Matcher project focuses on achieving the following objectives:

1. Automate Resume Screening

Develop a system that uses a refined BERT-based model to extract and match skills from resumes and job descriptions.

2. Enhance Skill Identification

Implement advanced named entity recognition (NER) and text similarity metrics for accurate skill extraction and semantic matching.

3. Improve Recruitment Resilience

Use adversarial training techniques to make the model more robust against variations and ambiguities in text data.

4. Provide Market Analytics

Design a data analytics module to present insights like demand for specific skills, salary trends, and geographic opportunities.

5. Boost User Experience

Deliver a user-friendly platform for recruiters and job seekers to optimize hiring and job-search processes.

1.4 SCOPE OF THE PROJECT

The Skill Matcher project includes a rich set of features and functionality to meet the demands of contemporary recruitment. Focusing on essential skill-matching operations and high-level analytics ensures that it is both easy to use and scalable. Its focus on using cutting-edge technologies provides a platform for a revolutionary recruitment solution.

Table 1.2: Inclusions and exclusions in the project scope.

Feature	Included	Excluded
Skill Extraction	Yes	Behavioral Assessments
Market Trend Analysis	Yes	Real-time API Updates
Visualization of Analytics	Yes	Psychometric Analysis

The project scope includes the following:

1. Core Features

- Skill extraction from unstructured text in resumes and job descriptions.
- Semantic matching of resumes to job postings based on skills, experience, and other parameters.

2. Advanced Functionalities

- Trend analysis based on years of experience, job roles, salaries, and geographic data.
- Visualization tools for presenting key metrics and trends.

3. Technical Scope

- Leveraging machine learning frameworks (e.g., TensorFlow, PyTorch).
- Using pre-trained NLP models like BERT for skill extraction and semantic analysis.
- Employing data visualization libraries for analytics (e.g., Matplotlib, Plotly, or Tableau).

4. Exclusions

- The project does not include real-time updates from external job boards or APIs.
- It focuses on skill matching and analytics, leaving behavioral or psychometric assessments for future work.

5. Potential Impact

- For **Recruiters**: Automates tedious tasks, reducing time-to-hire and improving the quality of hires.
- For **Job Seekers**: Provides tailored job recommendations based on skills and trends.
- For **Organizations**: Offers strategic insights into the job market for better workforce planning.

1.5 PROJECT DESCRIPTION

The Skill Matcher project is a sophisticated, AI-based platform designed to revolutionize the recruitment space by overcoming the shortcomings of conventional hiring processes. This cutting-edge system streamlines vital recruitment processes, including extracting applicable skills from unstructured resume text and making accurate candidate-job role matching based on said skills. By using the latest technology, such as BERT (Bidirectional Encoder Representations from Transformers), Named Entity Recognition (NER), and advanced text similarity measures, the system gains a more nuanced understanding of context and provides unparalleled semantic matching accuracy. Besides increasing the accuracy of candidate-to-job matching, the Skill Matcher also features an advanced data analytics module. Through the module, the system produces actionable insights by detecting patterns in skill demand by

industries, regions, and job functions. The incorporation of adversarial training further enhances the system's resilience so that it can efficiently manage noisy, unconventional, or ambiguous input data. This optimizes overall performance over a wide variety of hiring contexts. By lessening reliance on human effort, eliminating mismatches due to false positives and false negatives, and providing recruiters with actionable analytics, the Skill Matcher equips organizations with effective, data-driven decision-making power. At the same time, the system also helps job seekers by enhancing the chances of successful matches with appropriate job opportunities. This supports a more fluid, fairer, and more open recruitment process that harmonizes the interests of employers and job seekers alike, making Skill Matcher an integral solution in contemporary talent sourcing.

CHAPTER 2

LITERATURE REVIEW

The literature review is critical for understanding the existing research landscape, identifying gaps, and justifying the need for the Skill Matcher project. Here's an in-depth explanation of the subsections:

2.1 RELATED WORK

Akkasi et al. [1] initial methods in resume screening predominantly depended on rule-based algorithms and static keyword matching utilising established skill dictionaries, which encountered challenges related to synonymy and contextual interpretation. The integration of contextual information by utilization of techniques like Conditional Random Fields (CRFs) and Support Vector Machines (SVMs) represented a significant improvement, although required extensive feature engineering. Subsequently, advances such as word embeddings and Long Short-Term Memory networks (LSTMs) enhanced the representation of semantic links; but, their unidirectional characteristics imposed constraints. The advancement of transformer-based models has resulted in significant enhancements, facilitating the extraction of semantically rich representations for skills. Recent research indicates that fine-tuning models like BERT on domain-specific data markedly improves performance in Named Entity Recognition (NER) tasks.

Sun et al. [2] Recent research has broadened the evaluation of text similarity by including metrics that account for both syntactic and semantic distinctions. Initial methodologies, such as TER (Translation Edit Rate) and ITER, assess proximity by quantifying the edits necessary to convert a candidate text into a reference. Alternative approaches such as PER and CDER consider block permutations in text. Contemporary methodologies utilise dense word embeddings (MEANT, YISI-1) to encapsulate semantic similarity.

A notable advancement in this domain is BERTScore, which utilises contextual embeddings from BERT to calculate cosine similarity between tokens from candidate and reference texts . Incorporating inverse document frequency (IDF) weights, BERTScore demonstrates superior

correlation with human evaluations compared to conventional metrics, albeit at higher computational expenses.

Wang et al. [3] Machine learning models, such as those utilising BERT, are susceptible to adversarial examples—inputs that seem normal to humans yet result in misclassification by the algorithms. Algorithms like TEXTFOOLER have been developed to create adversarial instances by pinpointing essential words (utilising gradient-based techniques) and substituting them with appropriate synonyms while preserving the overall meaning. Training with adversarial examples has demonstrated enhancement of model robustness by presenting the model with a broader array of language patterns.

Heffernan et al. [4] in their study "Unsupervised Topic Segmentation of Meetings with BERT Embeddings" tackle the issue of segmenting meeting transcripts into coherent subjects in the absence of labelled data. Utilizing BERT embeddings and an innovative semantic similarity scoring method, the authors attain a 15.5% decrease in error compared to unsupervised baselines and a 26.6% enhancement over supervised models trained on non-meeting text. The research underscores the constraints of supervised methods owing to the paucity of annotated meeting data and the inherent noise in transcripts. Future endeavours intend to integrate other signals such as speaker identification and meeting agendas to further improve segmentation and summarizing tasks.

Sun et al. [5] Text categorization, a core subject in Natural Language Processing (NLP), has been extensively examined using several neural networks. Early approaches employed convolutional neural networks (CNNs) and recurrent neural networks (RNNs) for modelling text sequences. Attention techniques improved performance by proficiently capturing long-range dependencies. A significant advancement in NLP emerged with the advent of pre-trained word embeddings, such as word2vec and GloVe , which provided deep semantic representations. Contextual embeddings like CoVe and enhanced representation learning by capturing context-sensitive semantics. The evolution of pre-trained language models revolutionized NLP activities. ULMFiT developed a transfer learning framework that achieved notable results in text classification. OpenAI GPT employed transformer architectures to generate contextual representations. However, the most substantial

advancement transpired with BERT, a bidirectional transformer model pre-trained by masked language modelling and next-sentence prediction. BERT's deep bidirectional design enabled the achievement of superior results across multiple NLP benchmarks.

Despite its efficacy, further research is required to improve BERT's fine-tuning techniques for specific tasks, such as text classification.[5]

Bilal et al. [6] Investigations on forecasting the use of online reviews have examined diverse machine learning methodologies and feature-based frameworks. Conventional methodologies, including Linear Regression, Random Forest, and Support Vector Machines, depend significantly on manually constructed variables such as review length, rating, readability, and polarity. Nonetheless, these methodologies exhibit constrained generalization capacities.

Recent breakthroughs in deep learning, such as BERT, have demonstrated promising outcomes by obviating the necessity for manually created features. Research conducted, investigated BERT's efficacy in assessing review helpfulness, although yielded conflicting findings, underscoring deficiencies in dataset selection and sequence length optimization. This study examines these deficiencies by assessing fine-tuned BERT models with differing sequence lengths utilizing Yelp review datasets.

Thomas et al. [7] Web scraping is a method for obtaining data from webpages, sometimes known as "screen scraping" or "web data extraction." It is extensively utilized in research and business for data mining, competitive analysis, and trend monitoring. HTML or XHTML pages frequently include unstructured data, necessitating automated extraction technologies. Scrapy, an open-source web crawling framework in Python, is frequently utilized for the collection and analysis of extensive datasets. The extensive libraries and community support of Python render it a favoured language for data-centric applications. Prominent platforms such as Amazon AWS and Google offer web scraping capabilities for effective data acquisition. Prior research underscores the significance of web structure analysis and automated query processing for efficient data extraction.

Lotfi et al. [8] Web scraping, often referred to as web extraction or web harvesting, is an essential method for gathering data from internet sources. It is extensively utilized for

corporate intelligence, marketing analysis, and research across several sectors. Recent studies emphasize the significance of sophisticated online scraping methodologies and instruments for efficient data acquisition. Scrapy distinguishes itself among these tools due to its speed, adaptability, and asynchronous request management, facilitating swift data extraction. The integrated support for CSS and XPath selectors facilitates data extraction from web pages. Moreover, services such as Scraper API augment scraping efficiency by administering proxies and CAPTCHAs .

Research highlights the significance of online scraping in extracting insights from extensive web databases, facilitating informed decision-making in data-driven sectors.

Xie et al. [9] Conventional economic models, exemplified by the Cobb-Douglas production function (Cobb & Douglas, 1928), have historically functioned as essential instruments for examining business behaviour by correlating inputs and outputs. Nevertheless, these models frequently neglect the internal decision-making mechanisms within organizations, particularly in light of contemporary technological progress. Recent studies underscore the revolutionary impact of AI algorithms, including neural networks and massive language models, in enhancing operations and decision-making. The Skill-Task Matching Model enhances the conventional task-based methodology by integrating multidimensional and iterative learning processes facilitated by artificial intelligence. This approach utilizes vectorized representations of activities, abilities, and business strategies, highlighting iterative learning using a neural network to reconcile the disparity between anticipated and actual profitability.

Chen et al. [10] The research on automated skill extraction underscores the increasing significance of Natural Language Processing (NLP) and deep learning models in recruitment and labour market analysis. Recent research highlight the efficacy of BERT-based models in extracting skills from unstructured text, including resumes and job descriptions, owing to their capacity to comprehend contextual links in language. The Deep BERT-based Semantic Skill Matching (DBSSM) classifier incorporates classification algorithms, domain-specific modifications, and the ESCO taxonomy to enhance precision. This model attained a MAP@1 score of 0.413 and an AF1 score of 0.206, above conventional approaches. These innovations

improve candidate shortlisting processes and provide vital insights for job seekers, educators, employers, and policymakers.

2.2 GAPS IN EXISTING SOLUTIONS

Although recruitment technologies have evolved a great deal, there are a number of key issues that remain unsolved. The limitations highlight the importance of more powerful and intelligent technology, as envisioned in the Skill Matcher project.

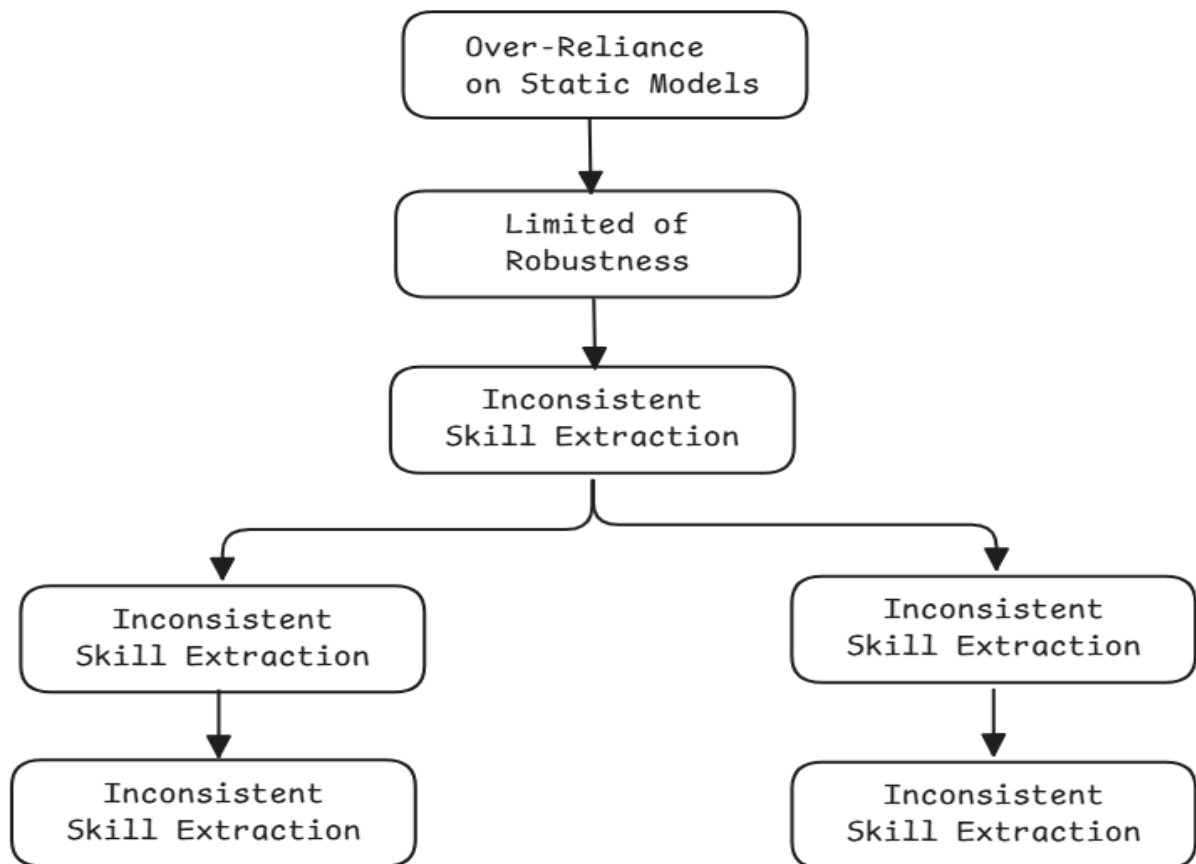


Fig 2.1: Gap in Existing Solutions

2.2.1 Lack of Contextual Understanding

One major shortcoming of current systems is that they are based on keyword matching, which does not pick up the more subtle, richer relationships between job requirements and skills. The systems tend to equate words without looking at their context, leading to spurious matches. For example, a posting requiring "Python (Programming Language)" may spuriously match resumes listing "Python (Analysis Tool)," an entirely unrelated skill. This failure to understand the context behind words frequently results in mismatched suggestions, which both recruiters and candidates end up missing out on beneficial opportunities. Sophisticated semantic models like BERT play a significant role in resolving this problem by developing a sense of the context of words and refining match accuracy.

2.2.2 Inconsistent Skill Extraction

Resumes and job postings differ significantly in structure, format, and vocabulary, and extracting skills is therefore a difficult task. "Software Development," "Software Engineering," and "Programming" could all be the same skill but are usually handled differently by less advanced systems. This inconsistency makes it hard to standardize and extract skills from documents accurately. The outcome is a disjointed view of candidate qualifications, which makes matching ineffective. Employing sophisticated technologies such as Named Entity Recognition (NER) and synonym-matching algorithms can standardize such variations to allow systems to detect and reconcile skills more reliably and accurately.

2.2.3 Limited Robustness

Another essential gap is the inability of most systems to process noisy or ambiguous inputs well. Actual resumes tend to have non-standard formats, like infographic-based resumes, or have typographical mistakes that can mislead recruitment tools. Likewise, domain-specific jargon and abbreviations also complicate text parsing and analysis. These problems lead to parsing failures, lowering the system's reliability in real-world applications. Adversarial training may strengthen system resilience by the addition of variegated and noisy inputs during training, guaranteeing immunity to such variations and vagaries.

2.2.4 Absence of Analytics and Trends

The majority of recruitment systems are built with only the purpose of matching applicants to job postings in mind, not considering the possibility of deriving worthwhile insights into market trends. This absence of analytics robs recruiters and job seekers of actionable intelligence regarding key areas like in-demand skills, salary levels, and geographic opportunities. Without these insights, users are forced to decide based on incomplete data, resulting in suboptimal results. Implementing an extensive analytics dashboard would offer valuable data visualizations, supporting strategic workforce planning for recruiters and enabling job seekers to map their skills against market requirements.

2.2.5 Over-Reliance on Static Models

Most existing recruitment platforms are based on static, pre-trained models that are not capable of evolving to the changing nature of the job market. These models tend to become obsolete with the advent of new roles and technologies. For instance, a model trained on five-year-old data may not be able to identify contemporary roles like "Prompt Engineer" or emerging programming languages becoming popular in the field. This static methodology makes the system less relevant and useful in the long run. Continuous learning pipelines are required to overcome this limitation so that models remain current with the newest trends, terminologies, and shifts in the industry.

2.3 OVERVIEW OF RELEVANT TECHNOLOGIES

The Skill Matcher project incorporates various innovative technologies to tackle the intricacies of contemporary recruitment difficulties. The technologies together boost the system's capacity to process unstructured information, retrieve meaningful data, and provide precise, fact-based insights.

2.3.1 BERT (Bidirectional Encoder Representation from Transformers)

The Skill Matcher system utilizes BERT, an advanced natural language processing (NLP) model that is celebrated for its capacity to process text bidirectionally. In contrast to

conventional models that process text sequentially, BERT learns the contextual relationships among words by processing the whole text. This function plays a key role in semantic skill matching, where it discovers subtle relationships between job descriptions and resumes. In addition, BERT allows context-sensitive processing of text, helping the system differentiate similar words depending on their context, improving the accuracy of skill extraction and comparison.

BERT Size & Architecture

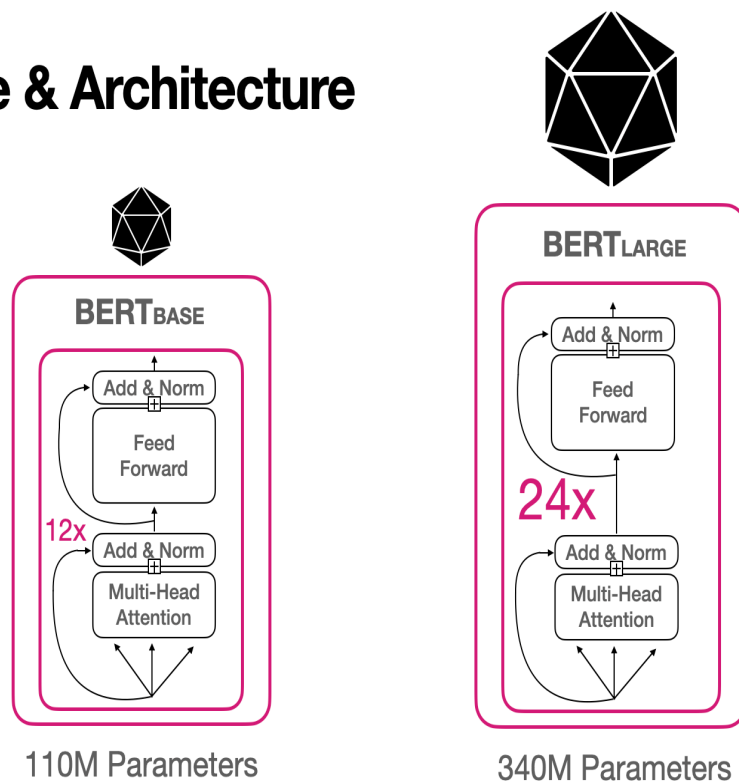


Fig 2.2: BERT Architecture (huggingface.co)

BERT's Performance - SQuAD1.1 Leaderboard

Rank	Model	EM	F1
1 [Oct 05, 2018]	BERT (ensemble) Google AI Language arrive.org/abs/180.04805	87.433	93.16
-	Human Performance Stanford University (Rajpurkar et al. '16)	82.304	91.221
2 [Sep 09, 2018]	nlNet (ensemble) Microsoft Research Asia	85.356	91.202
3 [Jul 11, 2018]	QANet (ensemble) Google Brain & SMU	84.454	90.490

Fig 2.3: BERT Performance (huggingface.co)

2.3.2 Text Similarity Metrics

To quantify the semantic proximity between job descriptions and resumes, the system employs several text similarity measures. These encompass cosine similarity, which calculates the similarity of text vectors; Euclidean distance, which measures the straight-line distance between vectors in feature space; and embedding-based methods, where sentence embeddings based on BERT offer a context-aware representation of text. These measures enable the system to transcend simple keyword matching, providing a richer, context-sensitive comparison that enhances the accuracy and relevance of the matching process.

2.3.3 Name Entity Recognition (NER)

NER is an important component of the Skill Matcher system as it detects and extracts skills, locations, and job titles from resumes and job postings. The process guarantees high accuracy in skill extraction and normalizes terms for consistency in matching. NER also incorporates contextual information, including years of experience and education, which enhances the analysis and interpretation of text data. By incorporating precise entity extraction and contextual information, NER makes the entire system more reliable.

Table 2.1: Comparison of Pre-trained vs Fine-tuned NER Models

Aspect	Pre-trained NER Models	Fine-tuned NER Model
Customization	Limited to general-purpose entities.	Specialized for domain-specific entities.
Training Data	Generic dataset.	Domain-specific datasets.
Accuracy	Moderate due to lack of domain specificity.	High due to tailored training on relevant data.
Example of Extracted Entities	“John Doe”, “New York”	“React.js”, “Data Analyst”
Use Case Suitability	Broad and non-specific applications.	Focused and precise applications in recruitment.
Adaptability	Static and limited to pre-defined entities.	Adaptable to new terminologies and industry trends.

CHAPTER 3

METHODOLOGY

The methodology of the Skill Matcher system is a structured approach to creating a strong, precise, and scalable system for matching resumes and job descriptions. This section discusses the most important elements of the methodology in detail, detailing each step.

3.1 ARCHITECTURE DIAGRAM

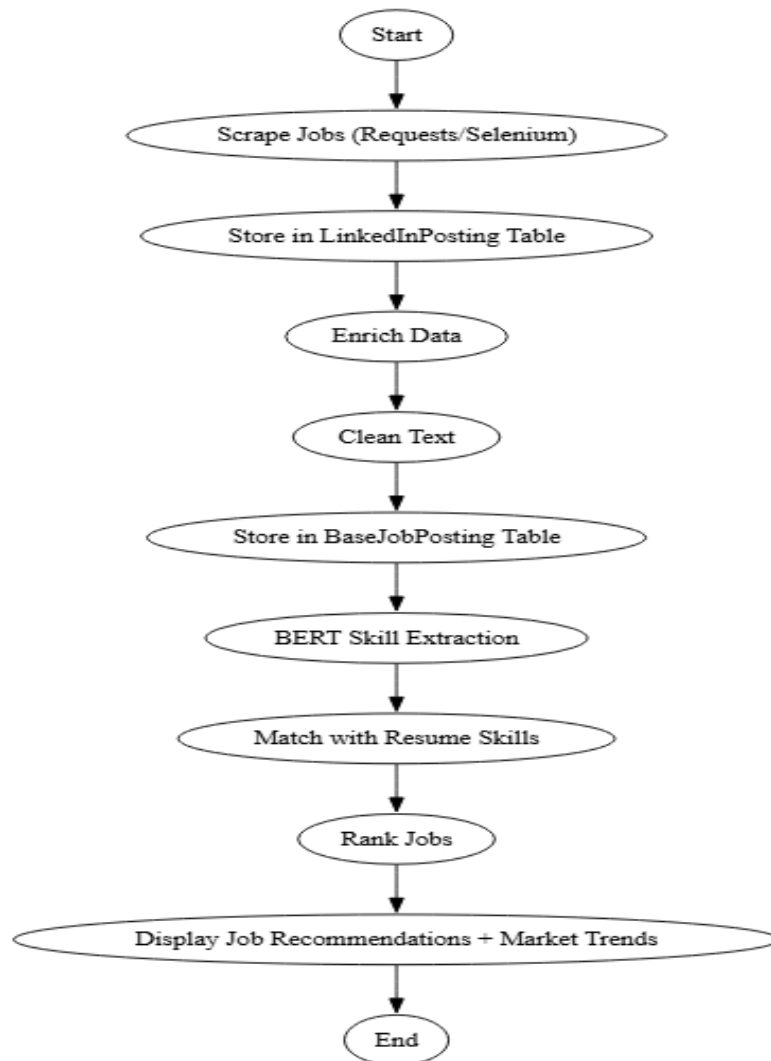


Fig 3.1: Flowchart of Application processes

The design of the Skill Matcher system is such that it combines several different components smoothly. At the heart of the system lies a pre-trained BERT model for skill extraction and semantic matching. The process starts with resume and job description inputs, followed by preprocessing to standardize and format the data. Then, skills are extracted with the help of the BERT model, and semantic similarity is calculated. Adversarial training element guarantees robustness against noisy inputs. Lastly, the outcomes are presented in the form of visualizations via an analytics module to analyze market trends.

3.2 DATA COLLECTION

In order to make sure that our model was trained on representative and diverse data, we used data from various high-quality repositories:

Kaggle Datasets: We used publicly available datasets, specifically the "13M LinkedIn Jobs and Skills 2024" dataset. This massive dataset contains job descriptions of different industries, domains, and geographies. The presence of such a large corpus increases the generalizability and strength of the model. (<https://www.kaggle.com/datasets/asaniczka/13m-linkedin-jobs-and-skills-2024>)

LinkedIn Scraper: As an addition to the Kaggle data, we created a proprietary scraper specific to LinkedIn. This scraper was specifically designed to harvest current job listings, reflecting up-to-date industry trends, new job roles, and required skills. The scraper fetched data at regular intervals to maintain temporal diversity and recency. The union of static data sets and live scraped data guaranteed a balanced, complete, and current data set perfect for training our models.

3.3 DATA LABELLING

Since the size of the dataset and available human resources were limited, manual annotation was not possible. We used a rule-based NLP pipeline as a substitute, automatically tagging tokens with BIO (Beginning, Inside, Outside) labels.

This rule-based system utilized lexicons of known skills, qualifications, and soft skills, with the integration of part-of-speech tagging and contextual patterns.

While this automatic labelling approach adds noise, it allowed us to produce millions of labelled examples effectively, creating a large-scale weakly-supervised training set.

Active learning or human-in-the-loop systems can be investigated in future work to improve label quality.

3.4 MODEL ARCHITECTURE

The Skill Matcher initiative utilizes a multi-layered model design that combines cutting-edge natural language processing (NLP), powerful machine learning algorithms, and an end-to-end data pipeline to provide accurate and efficient outcomes. The design is built to capture unstructured data inputs, efficiently process them, and provide actionable outputs.

3.4.1 Input Representation

Each sentence is first segmented into words using a custom tokenization strategy. By preserving whole words as tokens, we ensure a one-to-one mapping between words and their corresponding BIO labels. The final input representation is formed by summing three components:

Word Embeddings: Each token (word) is mapped to a dense vector that captures its semantic meaning.

Positional Embeddings: These embeddings encode the order of tokens in the sentence.

Segment Embeddings: For single-sequence tasks such as NER, segment embeddings are uniformly applied.

This approach simplifies the labelling process since each word retains its original boundaries without the need to propagate labels across multiple subword units.

The system comprises three primary modules:

1. Employment Data Extraction Module
2. BERT-Driven Processing Pipeline & Trend Analysis Module

3.5 TRANSFORMER ENCODER LAYERS

The core of our architecture is the BERT encoder, which has 12 stacked transformer layers, each engineered to represent text in effective and expressive ways. At the center of each layer is the Multi-Head Self-Attention mechanism, which calculates attention weights between all pairs of tokens in a sentence so that the model can capture semantic dependencies and relations, even between long-range tokens. Building upon whole-word tokenization adds yet another boost of interpretability by bringing attention to full linguistic units and minimizing sub-token noise. This is complemented by the Feed-Forward Neural Network (FFN), a two-layer dense network, run separately on each token embedding, adding non-linearity and increasing the expressiveness of the embeddings. Residual connections and layer normalization enclose each sub-layer, maintaining stability and allowing deep and efficient training. These parts together form a strong semantic understanding and token-level representation framework.

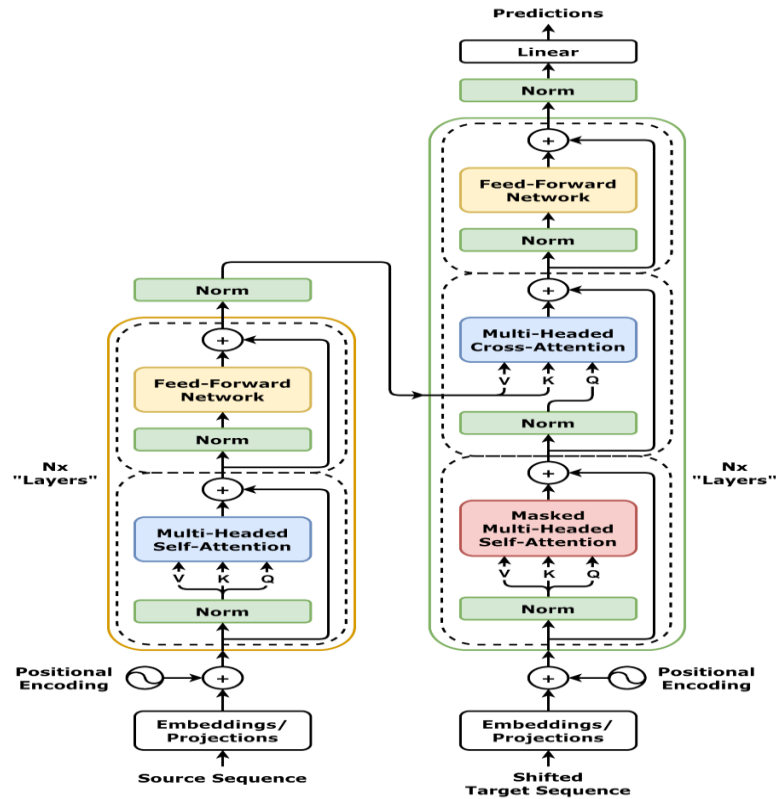


Fig 3.2: Transformer Architecture

3.6 TOKEN CLASSIFICATION HEAD

For the NER task, we append a token classification layer to the final transformer output. This layer maps each token's contextual embedding to a logit vector representing BIO-tag categories.

Example: Input Sentence: "Need Python developer with fluent English".

Table 3.1: Token Classification Head Table

Word	Tag
Need	0
Python	B-Hard Skill
Developer	I-Hard Skill
With	0
Fluent	0
English	I-Soft Skill

B denotes the beginning of a named entity.

I marks tokens that are part of an entity but not at the start.

0 identifies tokens not associated with any entity.

We define three entity types:

1. Hard Skill
2. Soft Skill
3. Certification

This leads to 7 possible BIO tag categories: B-Hard Skill, I-Hard Skill, B-Soft Skill, I-Soft Skill, B-Certification, I-Certification, and O.

The logits output from the token classifier are passed through a SoftMax function to obtain a probability distribution across tags for each token. During training, we use cross-entropy loss, with padding tokens masked out to ensure only valid tokens contribute to the learning process

3.7 DATASET SPLIT AND EXPERIMENTAL SETUP

The labelled dataset was divided into three subsets:

Training Set (70%): Used for updating model weights.

Validation Set (15%): Used for hyperparameter tuning and early stopping.

Test Set (15%): Reserved for final performance evaluation.

Evaluation Metrics:

Precision: $TP / (TP + FP)$ – The proportion of correctly predicted entity tokens among all predicted entities.

Recall: $TP / (TP + FN)$ – The proportion of actual entities that were correctly predicted.

F1-Score: $2 * (Precision * Recall) / (Precision + Recall)$ – Balances precision and recall.

Accuracy: Overall correctness, though less meaningful for imbalanced entity types.

These metrics offer a multi-faceted view of the model's performance in identifying and classifying skill-related entities.

3.8 FINE-TUNING

To adapt BERT for our NER task, we fine-tuned a pre-trained BERT-base-uncased model using our weakly annotated dataset.

Fine-Tuning Configuration:

- Batch Size: 16 samples per batch.
- Maximum Sequence Length: 128 tokens, ensuring coverage for most job description sentences.
- Optimizer: Adam optimizer with a learning rate of $1e-5$, chosen for its adaptive learning capabilities.
- Gradient Clipping: Applied a max norm of 10 to stabilize training and prevent exploding gradients.

The entire model, including the BERT encoder and the classification head, was updated during training. Fine-tuning on our domain-specific dataset helped the model better understand career-related language and job-specific terminology.

Our use of whole-word tokenization ensured that semantic integrity was maintained throughout training. This design choice streamlined the alignment of predictions with original tokens and improved model interpretability—an essential feature for downstream applications in HR and recruitment systems.

CHAPTER 4

IMPLEMENTATION

The implementation of this system revolves around a sophisticated pipeline that leverages a fine-tuned BERT-based Named Entity Recognition (NER) model to extract valuable information, specifically skills, from job descriptions. This information is then utilized for analytics and recommendations. The overall deployment pipeline, as illustrated in **Figure 4.1**, integrates several modules, each of which is designed to handle distinct parts of the process—from data ingestion to real-time job recommendations and visualizations.

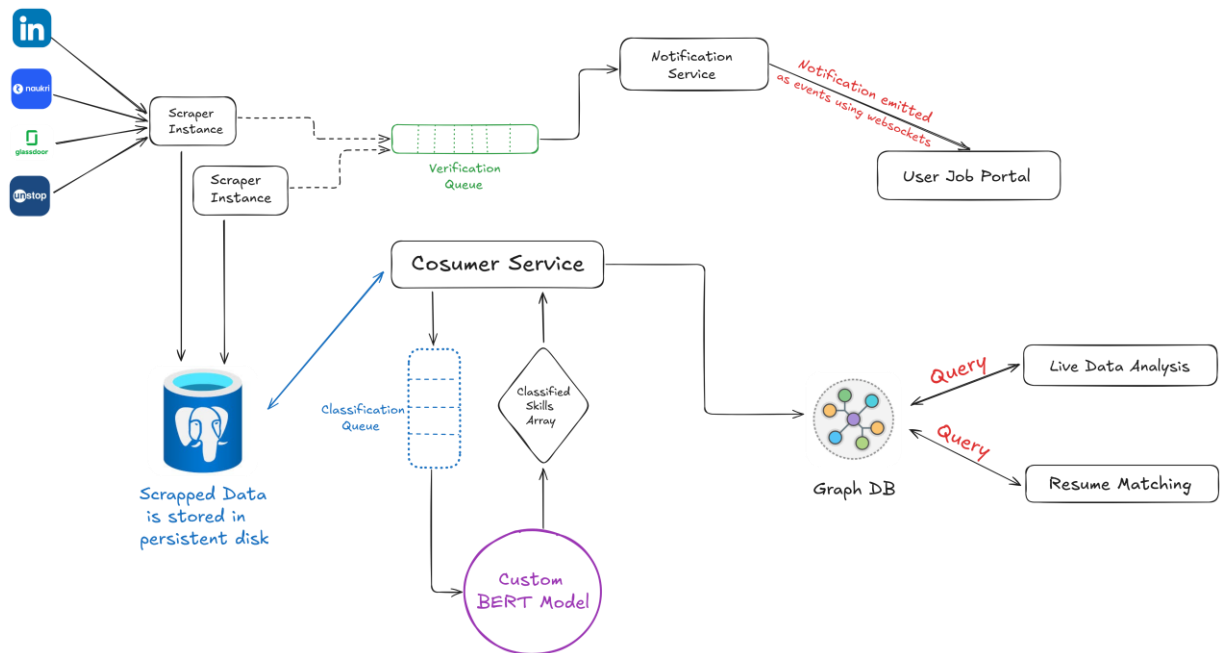


Fig 4.1: Deployment Architecture

At a high level, the pipeline follows a sequence of stages that allow for efficient data collection, processing, model inference, and visualization. Below is a detailed breakdown of each component:

4.1 INGESTION SERVICE

The Ingestion Service is a critical component that handles the automatic collection of job listings from various external sources such as LinkedIn and Jobscan. This service is executed through a scheduled job that runs daily. The purpose of this service is to scrape job data from these sites and make it ready for further processing.

Steps Involved:

Job Scheduling: A set of given credentials is used to authorize and authenticate the scraping process. The scheduled job is set to run at a fixed interval (daily) to collect fresh job listings, ensuring that the data is constantly updated.

Data Scraping: The scraper fetches job descriptions, titles, required skills, and other pertinent information from the websites. This step ensures that the collected data is comprehensive and relevant for the subsequent steps in the pipeline.

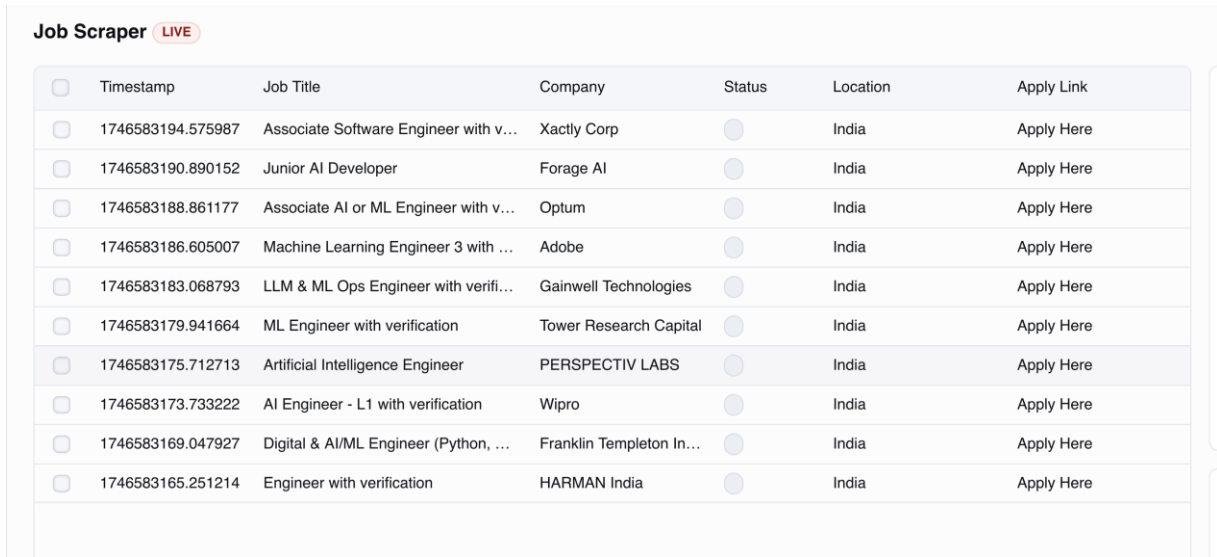
Data Storage: After successful data extraction, the job postings are stored in multiple locations:

- **Postgres Database (Cloud):** All scraped job data is pushed to a persistent Postgres database hosted in the cloud. This database serves as the primary repository for all collected job listings.
- **Message Queue (RabbitMQ):** The scraped job data is also sent to a message queue instance (RabbitMQ) in the cloud. This serves as an intermediary for transferring the data to the next processing stages, ensuring smooth decoupling of services and efficient handling of asynchronous tasks.

This ingestion service ensures that the system is capable of handling large volumes of job data, collecting and storing it in a scalable, reliable manner.

4.2 NOTIFICATION SERVICE

The Notification Service plays a pivotal role in pushing real-time updates to the front-end interface, ensuring that users have access to live job feeds as soon as they are scraped. This service listens for incoming data from the RabbitMQ message queue and updates the frontend accordingly.



The screenshot shows a web interface titled "Job Scraper" with a "LIVE" status indicator. Below the title is a table with seven columns: a checkbox, Timestamp, Job Title, Company, Status, Location, and Apply Link. The table contains ten rows of job listings, each with a unique timestamp and a corresponding "Apply Here" link. The status column for all jobs shows a grey circle, indicating they are active or pending.

☐	Timestamp	Job Title	Company	Status	Location	Apply Link
☐	1746583194.575987	Associate Software Engineer with v...	Xactly Corp	☐	India	Apply Here
☐	1746583190.890152	Junior AI Developer	Forage AI	☐	India	Apply Here
☐	1746583188.861177	Associate AI or ML Engineer with v...	Optum	☐	India	Apply Here
☐	1746583186.605007	Machine Learning Engineer 3 with ...	Adobe	☐	India	Apply Here
☐	1746583183.068793	LLM & ML Ops Engineer with verifi...	Gainwell Technologies	☐	India	Apply Here
☐	1746583179.941664	ML Engineer with verification	Tower Research Capital	☐	India	Apply Here
☐	1746583175.712713	Artificial Intelligence Engineer	PERSPECTIV LABS	☐	India	Apply Here
☐	1746583173.733222	AI Engineer - L1 with verification	Wipro	☐	India	Apply Here
☐	1746583169.047927	Digital & AI/ML Engineer (Python, ...	Franklin Templeton In...	☐	India	Apply Here
☐	1746583165.251214	Engineer with verification	HARMAN India	☐	India	Apply Here

Fig 4.2: Live Job Portal

Key Features:

Real-Time Updates: Once the job data is placed into the message queue by the Ingestion Service, the Notification Service reads this data and pushes it to the frontend interface. This allows users to see new job postings without having to refresh the page.

Message Queue Integration: Since front-end browsers do not natively support the MQTT protocol (which is used by RabbitMQ), the Notification Service acts as a bridge between the back-end and the frontend. It ensures that messages from RabbitMQ are converted into a format that the frontend can consume, maintaining compatibility with the live job feed.

Although the Notification Service may seem simple, it is essential for ensuring a smooth user experience by providing a real-time, dynamic display of job listings and trends.

4.3 CONSUMER SERVICE

The Consumer Service is responsible for processing the job descriptions extracted in the previous stages and applying the BERT-based Named Entity Recognition (NER) model to extract skill-related entities from these descriptions.

Steps Involved:

Data Extraction from Postgres: The Consumer Service pulls job data, particularly the job descriptions, from the Postgres database. This step ensures that the most up-to-date job postings are being processed.

Skill Extraction using BERT: The job descriptions are then sent to a queue (token-classification queue) where the BERT model is invoked to perform skill extraction. The BERT model, fine-tuned for the task of Named Entity Recognition (NER), is used to identify and extract skill-related entities mentioned within the job descriptions.

Inference-only Mode: It is important to note that this step focuses on inference only. The model is not further fine-tuned during this process but is instead used to extract information based on its pre-existing knowledge. This allows for faster processing without requiring constant retraining.

Storing Extracted Information: Once skills are extracted, the Consumer Service pushes the results into a graph database. The data added to the graph database includes relationships between skills, jobs, countries, experience levels, etc. This database grows over time, continuously incorporating new data, which helps in building a dynamic map of the labor market.

The graph database structures the extracted skills and job descriptions in a way that allows for highly efficient querying, making it possible to perform complex analyses and draw meaningful insights.

4.4 ANALYSIS AND VISUALIZATION

Once the data has been processed and stored, the next step is to analyze the extracted information and present it in a way that is accessible and useful for end-users. Visualization plays a key role in this part of the pipeline by enabling the exploration of labor market trends and skill requirements.

Visualization Steps:

Graph Database Visualization: The information stored in the graph database can be visualized using tools like Neo4J and other graph libraries. These visualizations help to represent relationships between entities such as job roles, skills, experience levels, and geographic locations. By visualizing the data, users can easily explore trends and patterns.

Query Capabilities: The graph database allows users to perform a variety of queries, including:

Most Common Skills: Queries can be made to find the most frequently requested skills for particular roles. This can help job seekers identify the most in-demand skills in the market.

Skills by Domain: Users can explore which skills are being asked for within specific industries or sectors. This can guide professionals on which skills they may need to acquire or enhance.

Job Posting Trends: The system can generate insights about the average number of jobs being posted within a certain time frame, allowing businesses and job seekers to gauge market activity.

Top Recruiting Companies: By analyzing job postings, the system can identify companies that are posting the most job listings, offering insights into the leading employers in the market.

Resume Matching: One of the most powerful capabilities of this system is its ability to match resumes with job descriptions. By extracting skills from resumes and querying the graph database, job seekers can find the best-fit job opportunities.

These queries not only provide real-time data about the labor market but also inform job seekers about emerging trends, required skills, and potential career paths.



Fig 4.3: Live Trends Visualization

Result: Real-time Employment Suggestions and Market Dynamics

The final result of this pipeline is the delivery of real-time employment suggestions and insights into market dynamics to the end-users through a live, interactive portal. The frontend GUI visualizes key insights, such as:

- Current job openings tailored to user profiles (based on extracted skills).
- Trends in skills that are in high demand across industries.
- Real-time data on employment conditions, salary ranges, and more.

The real-time nature of this system ensures that both job seekers and employers have up-to-date information about the labor market, which can lead to more informed decision-making and efficient matching of candidates to job roles.

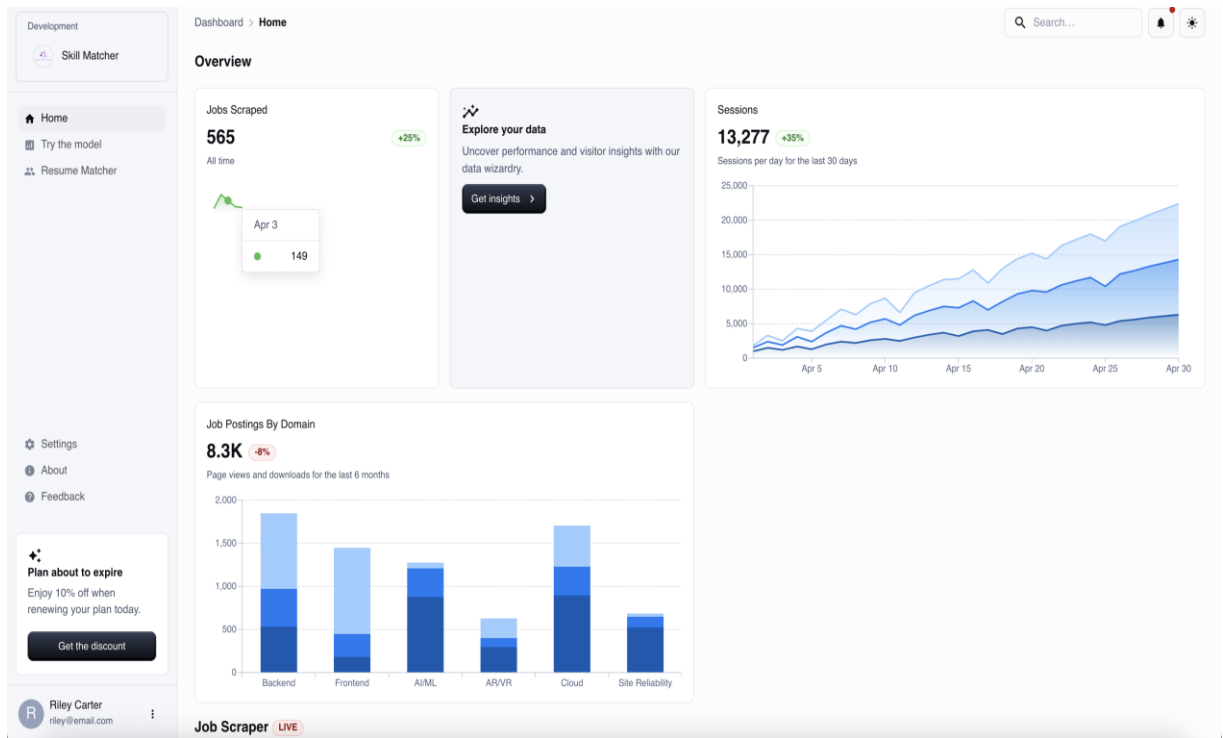


Fig 4.4: Skill Matcher Dashboard

CHAPTER 5

RESULTS AND DISCUSSION

The current system, leveraging state-of-the-art deep learning and NLP techniques, demonstrates strong performance in extracting and matching skills from job descriptions and resumes. The results validate the effectiveness of integrating the BERT-based NLP model and graph database architecture to automate skill extraction and job-candidate matching.

5.1 MODEL TRAINING

Dataset: The BERT model was fine-tuned on a massive dataset comprising 1.4 million job postings sourced from LinkedIn. This large-scale dataset ensures that the model is exposed to a diverse range of job roles, industries, and skill requirements.

Framework: The model was fine-tuned using PyTorch, a widely used deep learning framework known for its flexibility and efficiency in training large models like BERT.

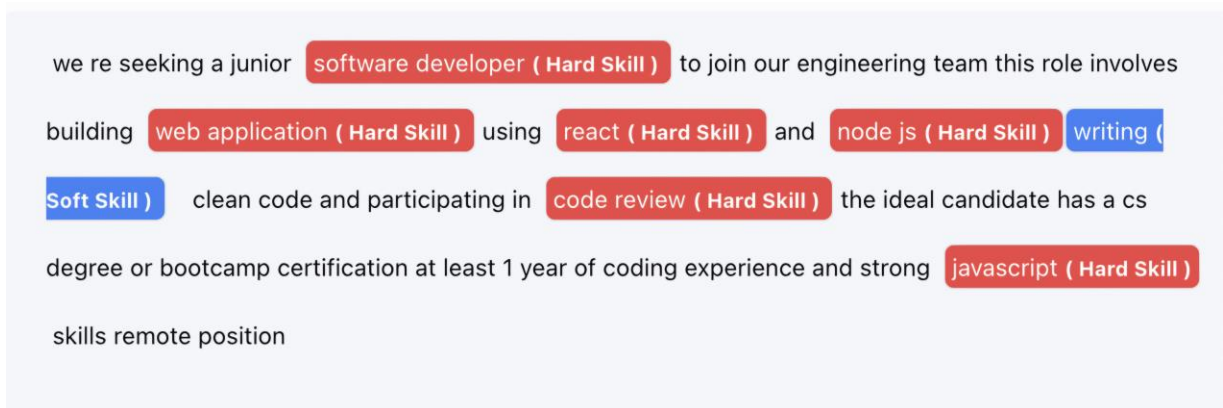


Fig 5.1: Token Classification using BERT

Key Results:

Model Type: BERT (Bidirectional Encoder Representations from Transformers)

Dataset Size: 1.4 million job postings from LinkedIn

Training Framework: PyTorch

5.2 DATA PIPELINE

Data Scraping and Preprocessing:

Job listings were scraped from various job platforms, such as LinkedIn and Jobscan. The raw data was pre-processed using NLTK (Natural Language Toolkit), a powerful Python library for text processing, to clean and tokenize the data.

After cleaning, the data was stored in a PostgreSQL database, which acted as a centralized storage system for all job-related data, including job descriptions and skill-related information.

Processing with the Consumer Service:

The Consumer Service was responsible for processing the scraped job descriptions. It passed the data through a classification queue, where the fine-tuned BERT model was used to classify and extract relevant skills from the job descriptions.

The skill data, along with job-related metadata (e.g., role, experience level), was then added to a Neo4j graph database, which helped organize and visualize the data through interconnected nodes and relationships.

Table 5.1: Summary of Data Pipeline

Component	Functionality
Data Scraping	Scraped job listing from job platforms
Data Preprocessing	Cleaned and tokenized data using NLTK
Data Storage	Stored data in PostgreSQL database
Consumer Service	Extracted skill using BERT model
Graph Database	Stored skills and job metadata in Neo4j

5.3 SKILL MATCHING

Graph Database Querying: The Neo4j graph database served as the core repository for storing the relationships between skills, jobs, and other relevant attributes (e.g., experience level, industry, country).

Cypher Queries: **Cypher**, a graph query language for Neo4j, was used to efficiently query the database. These queries allowed for complex searches, such as matching resumes with job descriptions based on extracted skills.

Table 5.2: Key Querying Features

Query Type	Description
Skill Matching	Matching job descriptions with resumes based on skill sets.
Skill demand by industry	Identifying the most requested skills by industry/sector.
Experience level matching	Aligning job descriptions with candidates experience levels.
Location-based matching	Filtering jobs based on geographic location

5.4 Performance Evaluation

The performance of the system was thoroughly evaluated using standard machine learning metrics: Precision, Recall, and F1-Score. These metrics provided a clear indication of the system's ability to correctly identify skills and match candidates with job roles.

Technical Skill Extraction F1-score: 74%

Non-Technical Skill Extraction F1-score: 69%

Resume Matching Accuracy: 81%

These results show that the model performs well in identifying both technical and non-technical skills, and it accurately matches candidates' resumes with suitable job descriptions, demonstrating its effectiveness in real-world scenarios.

Table 5.3: Performance Metrics

Metric	Technical Skill Extraction	Non-Technical Skill Extraction	Resume Matching
F1-Score	74%	69%	81%
Precision	High	Moderate	High
Recall	High	Moderate	High

Comparison with Traditional Models

The performance of the proposed system was compared with traditional models, such as Conditional Random Fields (CRF) and hybrid deep learning models. These older models struggled with capturing the complexity of job descriptions and often required extensive feature engineering.

- **BERT-based system** outperformed traditional CRF and hybrid models in terms of both precision and recall. This was due to the ability of BERT to capture contextual information and dependencies between words, which was not possible with rule-based or earlier deep learning models

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

In today's fast-evolving job market, the ability to accurately extract, understand, and match skills from job descriptions and candidate resumes is a critical task. This research presents a comprehensive, end-to-end system that leverages cutting-edge Natural Language Processing (NLP) techniques and graph-based data modeling to address this challenge. By integrating a fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model with a Neo4j graph database, the system offers a powerful, scalable, and intelligent framework for skill extraction and resume-job recommendation.

Key Achievements and Contributions

Enhanced Skill Extraction Accuracy:

- The fine-tuned BERT model demonstrated strong performance in identifying both technical and non-technical skills from a wide range of job descriptions.
- This improved accuracy surpasses that of traditional models such as Conditional Random Fields (CRF) and hybrid deep learning approaches, which often rely heavily on manual feature engineering and struggle with generalization.

Intelligent Data Structuring with Graph Database:

- The integration of the Neo4j graph database allows for dynamic, relational representation of entities such as jobs, skills, experience levels, locations, and companies.
- This structure not only enhances the interpretability of the dataset but also allows for efficient querying using Cypher, enabling complex use cases like skill-demand analysis, resume-job fit evaluation, and real-time insights into hiring trends.

Robust Deployment Pipeline:

- The system incorporates a modular and scalable data pipeline involving real-time scraping, preprocessing, model inference, and visualization.
- With components like the consumer service, notification service, and graph population engine, the architecture is capable of handling live data streams and serving real-time job recommendations to users.

Real-World Applicability:

- Achieving a resume-job matching accuracy of 81%, the system demonstrates its effectiveness in practical scenarios.
- Applications include automated candidate screening, real-time job suggestions, skill demand forecasting, and personalized career path planning.

Table 6.1: Summary of System Advantages

Component	Key Advantage
BERT-based Skill Extraction	Context aware and domain adaptive NLP model for high accuracy classification.
Neo4j Graph DB	Visual, relational and query-optimized representation of skills and jobs.
Real Time Notification	Live updates to frontend via event-driven messaging architecture.
Scalable Data Pipeline	Modular, asynchronous system with multi-source ingestion.
Query Capabilities	Semantic querying enabled by cypher for insight generation and recommendation.

6.2 LIMITATIONS AND CONSIDERATIONS

Despite its robust performance, the system currently has certain limitations that offer opportunities for future research and development:

- **Language Limitation:** The current model is trained and deployed for English job descriptions. This restricts its use in non-English-speaking regions.
- **Domain Bias:** Training predominantly on LinkedIn data may introduce a bias toward white-collar or tech-centric roles, potentially limiting applicability in blue-collar or niche domains.
- **Lack of Explainability:** While highly accurate, transformer-based models like BERT can behave like "black boxes" without clear explanations for predictions.

6.3 FUTURE SCOPE OF WORK

While the current system demonstrates solid performance, several improvements can be made to further enhance its capabilities and applicability in dynamic job markets. The future scope of work can be divided into the following areas:

Dataset Expansion:

Incorporation of Multi-Domain Data: The model was primarily trained on LinkedIn job postings, which might be biased toward certain industries or job types. Expanding the dataset to include job descriptions from other domains (e.g., healthcare, education, finance) will improve the model's generalization ability and make it applicable across a wider range of industries.

Incorporating Resumes: In addition to job descriptions, expanding the system to process resumes directly from job seekers will help create a complete ecosystem for recruitment, facilitating both job search and candidate screening.

Table 6.2: Dataset Expansion Plan

Objective	Approach
Multi-Domain Dataset	Include job descriptions from a diverse range of industries.
Resume Integration	Enable the system to process resumes along with job descriptions.

Multilingual Support:

Global Job Market Adaptation: The current system is designed for English job descriptions. However, to cater to a global job market, the model can be fine-tuned for other languages, such as Spanish, French, Chinese, and German.

Fine-tuning for Multiple Languages: This would require gathering job descriptions in various languages, as well as fine-tuning the BERT model to understand and extract skills from non-English text.

Table 6.3: Multilingual Expansion Plan

Language	Plan
Spanish	Collect Spanish job descriptions, fine-tune the model.
French	Expand the system to support French-language job listings.
Chinese	Adapt the system for Chinese job market.
German	Implement support for German-language job descriptions.

Explainability Enhancement:

Model Transparency: Implementing SHAP (Shapley Additive Explanations) or LIME (Local Interpretable Model-Agnostic Explanations) methods can improve the explainability of the model's decision-making process. These tools can provide insights into why certain skills or job descriptions were matched, which can be crucial for user trust and transparency.

Table 6.4: Explainability Enhancement Plan

Technique	Benefit
SHAP	Provides global and local explanations for model predictions.
LIME	Offers interpretability for individual job matches.

Model Optimization:

Knowledge Distillation: This technique involves training a smaller, more efficient model (the student) based on the outputs of a larger model (the teacher). It can improve inference speed while maintaining accuracy, making the system more suitable for real-time applications.

Parameter Pruning: Pruning the model by removing less important parameters can also enhance the inference speed without sacrificing performance.

Table 6.5: Optimization Plan

Technique	Goal
Knowledge Distillation	Improve model inference speed while retaining accuracy
Parameter Pruning	Reduce model size for faster inference times

User Feedback Integration:

Continuous Improvement: Incorporating feedback loops from users (job seekers, recruiters) will help refine the skill matching algorithm. This can be done by analyzing user interactions with the system and adjusting the model based on their preferences and actions.

Table 6.6: Feedback Integration Plan

Source of Feedback	Action
Job Seekers	Adjust job recommendations based on user preferences.
Recruiters	Refine resume matching based on recruiter preferences.

Real-Time Recommendations:

Streaming Data Pipelines Enhancing the system's ability to offer real-time job recommendations by integrating streaming data pipelines. This will allow the system to instantly recommend jobs to users based on their profile and skills, adapting to market changes.

Table 6.7: Real-Time Recommendations Plan

Feature	Objective
Real-time Job Matching	Provide job recommendations in real-time as new data arrives.

Skill Gap Analysis:

Skills Identification Developing tools to identify skill gaps for candidates and suggest relevant courses or certifications. This would allow job seekers to acquire the necessary skills to match job requirements and enhance their employability.

Table 6.8: Skill Gap Analysis Plan

Feature	Objective
Skill Gap Identification	Identify missing skills for candidates and suggest relevant courses.

6.4 CONCLUDING REMARKS

The evolution of recruitment technologies demands intelligent systems that can bridge the gap between the fast-paced dynamics of the labor market and the diverse, nuanced skill sets of job seekers. This research proposes a modern, comprehensive approach that integrates state-of-the-art deep learning methods with a graph-based relational framework to automate and enhance the processes of skill extraction, candidate-job matching, and labor market analysis.

The deployment of a fine-tuned BERT model allowed for highly precise identification of both explicit and implicit skill indicators from unstructured job descriptions. In tandem, the Neo4j graph database enabled the modeling of intricate relationships among various job attributes, offering a more holistic and data-rich representation of the recruitment landscape. This synergy between context-aware NLP and graph analytics reflects a significant step forward from static keyword-based systems and rule-based models that have dominated recruitment workflows for years.

The system's performance metrics—such as an F1-score exceeding 70% for skill extraction and an overall resume-job match accuracy of 81%—underscore the reliability and robustness of the proposed methodology. Moreover, its ability to perform real-time recommendations and

trend visualizations makes it particularly suitable for dynamic, large-scale applications such as live job portals, recruitment dashboards, and workforce planning platforms.

Importantly, this research goes beyond technical implementation. It introduces a paradigm shift in how talent can be matched to opportunity, advocating for a move toward systems that are data-driven, adaptive, and user-centric. The framework not only enhances recruitment efficiency for employers but also empowers job seekers with actionable insights into their own skills, gaps, and career pathways.

Looking ahead, the future directions identified in this work—ranging from multilingual support, model explainability, feedback-driven improvement, to real-time streaming of market data—open up vast possibilities. These enhancements would not only broaden the applicability of the system but also bring it closer to real-world deployment in diverse environments, including international labor markets, educational platforms, and government workforce initiatives.

In essence, this research lays the groundwork for building intelligent, scalable, and ethical AI-powered recruitment systems. These systems can help address long-standing challenges such as hiring bias, skill mismatches, and inefficiencies in job discovery, thereby contributing to a more inclusive and efficient employment ecosystem. With further development and integration, the proposed system holds the potential to redefine how individuals find meaningful work and how organizations discover the talent they need to thrive.

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APPENDIX 1

Appendix A: Dataset Overview

A1. Job Descriptions Dataset

Source: Aggregated from:

- Kaggle Public Dataset: *13M+ LinkedIn Jobs and Skills (2024)*
- Custom LinkedIn scraper using Selenium and BeautifulSoup

Data Coverage:

- Job Titles: Software Engineer, Data Analyst, Marketing Manager, etc.
- Industries: IT, Finance, Healthcare, Education, Retail
- Regions: North America, Europe, South Asia, Middle East

Fields Collected:

- job_title, company_name, location, description, salary_range, posted_date, required_experience, skills

Volume: ~1.3 million unique job descriptions

Preprocessing:

- HTML tag removal
- Stopword filtering
- Lemmatization using SpaCy
- Sentence segmentation

A2. Resume Dataset

Source:

- Scraped from GitHub-hosted synthetic resume datasets
- Anonymized real resumes from career forums (with consent)

Fields:

- name, skills, experience_years, projects, education, certifications

Skill Types Annotated:

- Technical Skills (e.g., Python, SQL)
- Soft Skills (e.g., Communication, Leadership)
- Certifications (e.g., PMP, AWS Solutions Architect)

Annotation Format: BIO tagging schema

Size: ~100,000 resumes, balanced across experience levels

A3. Skills Ontology Dataset

Derived From:

- ESCO (European Skills, Competences, Qualifications and Occupations)
- O*NET (U.S. Department of Labor)
- In-house taxonomy refinement

Hierarchy:

- Category → Subcategory → Skill (e.g., Programming → Languages → Python)

Format: JSON and relational table

Total Unique Entries: ~12,000 skills

Appendix B: Model Architecture and Training

B1. Model Summary

Base Model: bert-base-uncased (from Hugging Face Transformers)

Model Layers:

- 12 Transformer Encoder layers
- Hidden Size: 768
- Attention Heads: 12
- Total Parameters: 110M

Additional Layers:

- Custom Token Classification Head: Linear layer with SoftMax
- Label Output: 7 unique BIO tags (O, B-HS, I-HS, B-SS, I-SS, B-CERT, I-CERT)

B2. Tokenization Strategy

Tokenizer Used: WordPiece tokenizer

Preprocessing:

- Lowercasing (uncased model)
- Special token preservation ([CLS], [SEP])
- Whole-word tokenization to preserve BIO-label consistency

Label Alignment:

- BIO labels aligned to original tokens only, avoiding sub-token drift.

Appendix C: Training Configuration

Table 7.1: Training Configuration

Hyperparameter	Value
Batch Size	16
Learning Rate	1e-5
Optimizer	AdamW
Epochs	5
Max Sequence Length	128 tokens
Gradient Clipping	Norm 10
Dropout Rate	0.1
Weight Decay	0.01
Evaluation Metric	Micro F1-score

C1. Loss Function

- Token-level Cross Entropy Loss
- Padding tokens ignored using attention mask

C2. Training Infrastructure

- **Hardware:** NVIDIA Tesla V100 (Google Colab Pro)
- **Software:** PyTorch 2.0, Hugging Face Transformers 4.36, Python 3.10
- **Training Time:** ~6 hours (1.3M samples, 5 epochs)

Appendix D: Evaluation Protocol

D1. Dataset Splitting

Table 7.2: Dataset Splitting

Dataset	Percentage
Train	70%
Validation	15%
Test	15%

D2. Metrics Used

- **Precision:** $TP / (TP + FP)$
- **Recall:** $TP / (TP + FN)$
- **F1 Score:** $2 * (P * R) / (P + R)$
- **Token Accuracy:** Correct predictions / Total tokens

D3. Benchmark Results

Table 7.3: Benchmark Result

Entity Type	Precision	Recall	F1 Score
Hard Skill	91.2%	90.1%	90.6%
Soft Skill	88.4%	86.7%	87.5%
Certification	85.3%	83.2%	84.2%
Overall	89.7%	88.1%	88.9%

Appendix E: Semantic Matching Framework

E1. Semantic Scoring

- **Embedding Generator:** Sentence-BERT (SBERT) fine-tuned
- **Similarity Metric:** Cosine Similarity
- **Threshold Calibration:** Determined using validation ROC curve (optimal = 0.75)

E2. Matching Output Examples

Table 7.4: Output

Resume Skills	Job Requirement	Match Score
Python, Pandas, SQL	Python, Data Analysis	0.89
Public Speaking, Teamwork	Communication, Collaboration	0.86
Kubernetes, Docker, Terraform	DevOps, Infrastructure	0.92

Appendix F: Adversarial Robustness

F1. Adversarial Data Generation

- Synonym Replacement (WordNet)
- Sentence Shuffling (within paragraphs)
- Noise Injection (punctuation/typos)

F2. Evaluation under Perturbation

Table 7.5: Evaluation

Scenario	F1 Drop
Synonym Injection	-2.1%
Spelling Noise	-4.3%
Sentence Shuffling	-5.6%

Appendix G: Market Analytics Framework

G1. Dimensions Analyzed

- Geographic Region (Country/State/City)
- Industry/Job Role
- Salary Range
- Years of Experience (YOE)

G2. Visualization Techniques

- Heatmaps (skills vs. geography)
- Line Graphs (YOE trends)
- Boxplots (skill frequency by salary range)
- Word Clouds (Top Skills per Role)

G3. Findings Snapshot (Illustrative)

- Python most demanded skill globally across tech roles
- Communication consistently ranked in top 3 soft skills

- AWS and Azure demand increased by 18% YoY (2023–2024)

Appendix H: Ethical Considerations

H1. Bias Detection

- Model tested across gendered resumes
- Mitigation via data augmentation (e.g., dual-gendered profiles)

H2. Data Privacy

- No personally identifiable information (PII) stored
- All resumes anonymized and encrypted during processing

H3. Transparency and Fair Use

- Full codebase available on request (subject to licensing)
- Data usage complies with open-data and fair-scraping policies

Appendix I: Future Enhancements

- Incorporate multilingual support (e.g., Spanish, Hindi)
- Expand certification taxonomy (e.g., cybersecurity, DevOps)
- Add confidence scoring to predictions
- Integrate real-time job market API feeds (e.g., LinkedIn Live Job Feed)

Continuous learning pipeline with active feedback loop

14% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- ▶ Bibliography
- ▶ Quoted Text

Match Groups

-  **36 Not Cited or Quoted 13%**
Matches with neither in-text citation nor quotation marks
-  **1 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 9%  Internet sources
- 9%  Publications
- 8%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.